Drosophila cCRE Pipeline

Detailed implementation and scientific reasoning

drosophila-cCREs project

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Abstract

This document describes the implemented workflow that converts public ATAC-seq and ChIP-seq accessions into quality-reviewed biological-library BAMs, replicate-supported context peaks, a variable-width summit-aware master DNase-hypersensitive-site (DHS) registry, and a context-resolved regulatory element catalog. The downstream catalog combines ATAC accessibility and H3K27ac signal using assay-specific background TMM normalization, guarded two-component Gaussian mixture models, portable browser tracks, and IGV sessions. For the canonical dm6 atlas, the workflow also normalizes available Micro-C/Hi-C matrices, models contact decay where a context map is absent, and builds context-specific element–promoter and element–gene candidate tables. An integrated checksummed report summarizes all stages. The emphasis is both operational and scientific: each section states what the current rules do, why that choice was made, and which limitations remain. This is implementation documentation for the live `workflow/Snakefile` and its included rule files; it is not a substitute for inspecting the resolved configuration and provenance of a particular run.

Contents

1 Purpose and scope

The pipeline has three related scientific goals:

1. construct a stable atlas coordinate system—the master DHS registry— from reproducible ATAC accessibility evidence across the contexts named in the input table; and
2. annotate every master DHS in every context with accessibility, H3K27ac, a high-component posterior probability, TSS proximity, blacklist overlap, and a guarded high/low interpretation; and
3. use matched contact evidence and promoter activity to prioritize candidate genes for context-member elements without turning a ranking into a causal target assignment.

The public interface is accession-first. Users provide public accessions and biological metadata, not raw FASTQ paths. The canonical launcher is `src/run_pipeline.py`; it validates inputs, resolves defaults and accession layouts, writes a deterministic YAML configuration, and invokes the canonical `workflow/Snakefile`. Snakemake then owns dependency tracking, resource scheduling, rule environments, and restart behavior.

The current workflow includes ATAC-seq processing, TF and histone ChIP-seq processing, H3K27ac-based context activity annotation, catalog visualization, contact-map normalization, candidate-gene linking, and integrated reporting. It does not perform ABC candidate resizing or ABC scoring. It also does not import an external regulatory-element catalog or apply quantile normalization.

A master registry is derived from the project's own reviewed accessibility evidence before contacts are introduced. Contact information therefore annotates stable master coordinates but cannot change which DHSs exist. The link stage preserves observed contact, distance expectation, promoter activity, and separate rankings, making its evidential limits visible. ABC scoring remains outside this workflow because it would add a different candidate definition and calibration claim.

2 The data model

2.1 Run, library, context, and assay

Four identifiers have distinct meanings:

Accession or technical run
: One public run, or one run obtained by expanding a public experiment accession. Technical runs are processed independently through trimming and alignment.

Biological library
: The unit identified by `library_id`. Multiple technical runs may share one library ID. Their aligned BAMs are merged before duplicate marking so duplicates are defined across the complete biological library.

Context
: The tissue, stage, cell type, or other biological group in the sample-sheet `context` column.

Distinct ATAC libraries in one context are biological replicates. The same context label also links accepted ATAC and H3K27ac libraries during activity quantification.

Assay
One of `atac`, `h3k27ac`, `chip_tf`, or `chip_histone`. The `h3k27ac` label resolves to the histone ChIP processing branch but is retained as H3K27ac in the downstream atlas.

Collapsing these levels would create pseudoreplication or biased weights. A technical lane is not a biological replicate; a library should receive one QC decision; and a context estimate should give equal weight to accepted biological libraries rather than to the number of sequencing lanes.

2.2 Canonical sample sheet

The CSV or TSV input conforms to `schemas/sample-sheet.schema.yaml`. Four columns are required:

Column	Meaning
`accession`	Public run or experiment accession; experiments expand to their runs.
`library_id`	Biological library identifier; shared by technical runs only.
`assay`	Assay branch and its default peak behavior.
`context`	Replicate-pooling and downstream atlas group.

Optional fields specify treatment/control roles, an explicit `control_library`, MACS3 peak-calling parameters, adapters, alignment parameters, filtering parameters, and notes. ChIP input controls are explicit relationships. A blank control is a valid IP-only experiment and causes MACS3 to run without `-c`; it does not trigger an implicit control search.

Genome is a project-level launcher argument (`dm6` or `hg38`), and sequencing layout is resolved from accession metadata. Repeated per-row genome or raw FASTQ fields are deliberately absent from the public contract.

3 Staged execution and the manual checkpoint

3.1 Why there are separate invocations

The workflow is intentionally not one uninterrupted accession-to-catalog job. It is organized around immutable, checksummed artifacts and an explicit human review boundary:

Invocation 1: accessions → QC
download → trim → align → peaks and QC → pending final-BAM manifest and review table

⇓ **manual pass/fail review** ⇓

Invocation 2: reviewed final BAMs → master
validate → accepted ATAC replicate/context peaks → master DHS bundle

Invocation 3: accepted BAMs + frozen master → catalog/links/report
validate → count → normalize → annotate → normalize contacts → candidate links → tracks, IGV sessions, and report

There is no interactive pause inside Snakemake. The user copies the generated review table outside the result namespace, records one pass/fail decision for every library, and applies those decisions with `src/review_final_bam_manifest.py`. A master run rejects any pending ATAC decision. Quantification similarly requires accepted ATAC and H3K27ac artifacts; a rejected row must carry a reason.

An automatic pipeline can calculate metrics but cannot decide whether a library is biologically fit for a specific atlas. Making the review boundary an explicit file transformation preserves human judgment without hiding it in an interactive process or modifying generated manifests in place.

3.2 Input and stopping stages

Start	Required starting artifact	Allowed endpoints
`accessions`	Sample sheet plus resolved, checksummed download manifest	`trimming`, `alignment`, or `qc`
`trimming`	Checksummed trimming checkpoint	`trimming`, `alignment`, or `qc` in the source namespace
`alignment`	Checksummed alignment checkpoint or complete final-BAM manifest	`alignment` or `qc`
`qc`	QC checkpoint plus a reviewed ATAC final-BAM manifest for master construction	`qc` or `master`
`master`	Checksummed master bundle and accepted ATAC/H3K27ac BAM manifests when continuing	`master`, `quantification`, `catalog`, `links`, or `report`
`quantification`	Checksummed quantification checkpoint	`quantification`, `catalog`, `links`, or `report`
`catalog`	Checksummed catalog checkpoint, or an immutable catalog-bundle manifest	`catalog`, `links`, or `report` from a checkpoint; `links` only from a bundle manifest
`links`	Checksummed contact-link checkpoint	`links` or `report`

Start	Required starting artifact	Allowed endpoints
`report`	Checksummed report checkpoint	`report` validation

Each endpoint exports `provenance/manifests/<stage>.checkpoint.json`. It records the stage, scientific semantic digest, complete peak/refinement parameters where relevant, and SHA-256 hashes of the restart artifacts. Its stage-specific resolved configuration snapshot is retained below `provenance/configs/`, so advancing the same run does not invalidate an earlier boundary. Advancing `-until-stage` inside an allowed start mode changes only requested targets; it is excluded from the scientific semantic digest.

Strict reuse modes have no upstream fallback. For example, a quantification run cannot silently redownload or realign a missing library. It first verifies the declared files, SHA-256 hashes, genome, BAM/index coherence, coordinate compatibility, and filtering contract.

4 Reference preparation

For `dm6` and `hg38`, reference construction belongs to the workflow DAG. The resolved configuration records canonical FASTA, NCBI RefSeq GTF, and Blacklist v2 sources together with MD5 or SHA-256 checksums. The workflow:

1. downloads archives with resumable aria2 transfers and integrity checking;
2. decompresses the FASTA and blacklist through temporary files followed by atomic replacement;
3. builds the samtools faidx and Bowtie2 indexes;
4. derives chromosome sizes in FASTA order;
5. extracts one-base TSS records from the annotation; and
6. writes the configured autosome list used for background normalization.

The blacklist is retained as an annotation reference rather than used to discard alignments. At the catalog stage, master-DHS overlap with it is reported as a binary flag, an overlapping-base count, and a fraction of the master interval. The TSS BED supports ATAC TSS enrichment and regulatory-class annotation. The autosome list excludes sex chromosomes and mitochondrial DNA from the TMM background matrix.

Reference identity is part of the computation, not a machine-local prerequisite. Checksummed preparation in the DAG makes alignment, QC, normalization, and coordinates traceable to the same assembly and avoids silent reuse of a similarly named but different reference.

5 Acquisition, read QC, and trimming

5.1 Checksummed acquisition

Run and experiment accessions are normalized and experiments are expanded to their constituent runs. Acquisition produces immutable compressed FASTQs and a manifest with resolved paths, layout, and checksums. Resume state is preserved, and SRA-derived files are promoted only after conversion and compression complete. In the default `-backend auto` mode, runs whose direct ENA FASTQs fail transfer or integrity validation are converted with SRA Toolkit; ENA files from other runs that completed successfully remain in place. Explicit `-backend ena` mode disables this failover. The `-skip-download` path reuses an existing manifest after validation.

5.2 Raw and trimmed read assessment

Every technical run and mate receives raw FastQC. Reads are then trimmed with Cutadapt; defaults are Nextera adapters for ATAC and TruSeq adapters for ChIP, quality cutoff 20, minimum retained length 20 bp, error rate 0.1, and minimum adapter overlap 3 bp. Paired-end trimming is synchronized. FastQC is repeated on the trimmed output, and Cutadapt writes a JSON record used by MultiQC.

Running QC before and after trimming separates properties of the deposited reads from effects introduced by processing. Keeping technical runs separate at this stage reveals lane-specific failures before they are hidden by a library merge.

6 Alignment and biological-library BAM construction

6.1 Lane alignment

Each trimmed technical run is aligned independently with Bowtie2. The default preset is `-very-sensitive`. For paired-end data, the workflow also uses `-no-mixed`, `-no-discordant`, and a default maximum fragment length of 2,000 bp. Read-group IDs identify the technical run while the sample field identifies the biological library.

The streamed alignment is converted to BAM, name-sorted, processed with samtools fixmate -m, and coordinate-sorted. Temporary intermediates are checked with samtools quickcheck.

6.2 Merge, duplicate marking, and filtering

All lane BAMs sharing a biological library ID are merged. Duplicate marking is then performed once over the merged library. The final filter applies:

- minimum MAPQ 30 by default;
- exclusion of unmapped, secondary, QC-failed, and supplementary alignments;
- proper-pair and mapped-mate requirements for paired-end libraries;
- duplicate exclusion when `remove_duplicates=true`;
- mitochondrial-contig removal by default.

Blacklist-overlapping alignments are deliberately retained so that the same unfiltered signal supports peak calling and quantitative annotation. The resulting elements can then be removed, retained, or sensitivity-filtered by their recorded overlap rather than disappearing before inspection.

The result is a coordinate-sorted final BAM and BAI. Flagstat, stats, and idxstats summaries are calculated for every biological library. At the `alignment` boundary the workflow exports a checksummed final-BAM manifest; newly generated rows have `qc_status=pending_review`. They use the `short-read-processing-final-v2` contract, which records this blacklist-retaining behavior. A v1 BAM cannot be relabeled as v2 because reads removed by the old contract cannot be recovered; adopting v2 requires reprocessing from FASTQs. Master-bundle manifests record the input filtering contract as well, so a v1 master cannot silently enter a v2 catalog.

Duplicate marking before technical-run merging would miss molecules duplicated across lanes. Conversely, merging biological replicates would erase the replicate structure required for reproducibility checks. The chosen order therefore matches the experimental unit.

7 Peak calling

7.1 Default paired-end ATAC branch: Tn5/MACS3-qpois

The default ATAC implementation is explicitly insertion-based:

1. Retain proper pairs with nonzero template length and $0 < |\mathrm{TLEN}| < 150$ bp by default. Unmapped, mate-unmapped, secondary, QC-failed, duplicate, and supplementary records are excluded.
2. Apply the Tn5 offset with alignmentSieve –ATACshift.
3. Convert both shifted mates to separate one-base insertion records. A consistency check requires the shifted alignment count to equal the number of insertion records.
4. Run MACS3 callpeak on the BED input with `-f BED -nomodel -shift -75 -extsize 150 -keep-dup all -B -q 0.10`. The pileup and local lambda are deliberately unscaled; this branch must not use `-SPMR`.
5. Clip the treatment pileup and local lambda to chromosome bounds and calculate the qpois signal with macs3 bdgcmp -m qpois.
6. Refine candidate peaks over qpois exponents 2 through 325. Connected components shorter than 50 bp are rejected; components at most 400 bp are retained; wider components are followed to higher exponents so that they can split into narrower subcomponents. The default merge gap is one base.

This sequence runs for each accepted biological replicate and again on pooled insertions within each context. The pooled call provides context boundaries and signal; the replicate calls provide reproducibility evidence. The QC endpoint freezes each refined replicate BED together with the complete MACS3 and qpois-refinement parameter mappings. A QC-to-master invocation validates and reuses those lenient replicate calls, while still constructing the pooled context calls. It refuses reuse if the requested q-value, Tn5 geometry, exponent range, component lengths, or merge gap differs.

Two-ended Tn5 insertions represent cut sites more directly than treating an ATAC fragment as a ChIP enrichment interval. Computing qpois from unscaled pileup and lambda preserves MACS3's intended count-scale comparison. Progressive refinement constrains the geometry of broad candidates, but it is not an additional independent FDR-controlled test; candidate q-values remain the statistical starting point.

7.2 ChIP branches

ChIP treatment libraries use MACS3 with their explicit control BAM when one is declared and without `-c` for IP-only data. TF ChIP defaults to narrow peaks; histone ChIP, including H3K27ac, defaults to broad peaks. The default q-value is 0.01 and the default broad cutoff is 0.1. MACS3 format resolves from layout unless explicitly constrained.

Both narrow and broad ChIP calls request `-B -SPMR`, producing treatment-pileup and control-lambda bedGraphs in addition to the canonical narrowPeak or broadPeak file. Per-library ChIP CPM BigWigs are not generated because no downstream endpoint consumes them. The SPMR behavior is intentionally different from the ATAC qpois branch.

Narrow TF occupancy and broad histone enrichment have different expected geometries. Explicit control relationships prevent accidental cross-context matching, while allowing genuinely IP-only public experiments to remain representable and auditable.

8 Quality control and manual acceptance

The QC endpoint combines general and assay-specific evidence:

- raw/trimmed FastQC and Cutadapt metrics;
- final-BAM flagstat, stats, and idxstats;
- peak count and fraction of reads in peaks (FRiP);
- ATAC TSS profiles over ±2 kb, derived from temporary Tn5-shifted CPM tracks, and fragment-length histograms;
- ChIP treatment/control fingerprints when a control exists;
- phantompeakqualtools cross-correlation, fragment-shift estimates, NSC, RSC, and quality tags for ChIP treatments; and
- a MultiQC report and one consolidated review row per library.

For paired-end libraries, FRiP uses one valid same-chromosome fragment per read pair in the denominator and counts a fragment once if it overlaps any called peak. For single-end libraries, the unit is one filtered read. Thus FRiP is fragment-aware and is not inflated by counting two mates independently.

ATAC TSS enrichment is calculated from the aggregate 4-kb profile as the maximum signal in the central ±50-bp interval divided by the mean signal in the two terminal 100-bp flanks. The review table also reports the median ATAC fragment length, fraction below 150 bp, and fraction from 180–250 bp.

The user edits only `qc_decision`, `estimated_fragment_length_bp`, and `notes`. A failed row requires a reason. Single-end H3K27ac also requires a reviewed positive fragment length; the cross-correlation estimate is pre-populated as a suggestion rather than accepted silently.

No universal numeric cutoff is safe across all public datasets. The table puts depth, mapping, enrichment, fragment structure, and visual evidence next to the decision so inclusion criteria can be documented at library resolution.

9 Context peak reproducibility

Accepted ATAC libraries are grouped by the sample-sheet context. Their insertion BEDs are concatenated and a pooled qpois context call is made and refined. The pooled peaks are not accepted merely because they are present in the aggregate signal.

For every pooled peak P, each replicate peak set is reduced to an interval union and the coverage fraction is calculated:

$$f_r(P) = \frac{|P \cap U_r|}{|P|},$$

where U_r is the union of replicate r's peaks. A replicate supports the pooled peak when $f_r(P) \geq 0.5$ by default. The peak is retained when at least two biological libraries support it. Both thresholds are launcher parameters, and the minimum cannot exceed the accepted replicate count. The support table retains every pooled peak, all per-replicate fractions, the supporting library list, and the retain/reject decision.

Pooled data improve sensitivity and define a context-level signal, but pooling alone can promote a feature driven by one library. Requiring substantial coverage in multiple independent biological libraries makes reproducibility an explicit property of every retained context peak.

10 Summit-aware master DHS registry

The master registry is built only from replicate-supported context peaks. It is neither a simple interval union nor a fixed-width resizing operation.

10.1 Context summits

For each source peak, the workflow reads the pooled context BigWig and chooses the maximum-signal base. If the maximum occupies more than one plateau, it selects the midpoint of the plateau closest to the peak midpoint, with a leftmost deterministic tie break. If no finite signal is available, it uses the peak midpoint and records a summit fallback.

10.2 Cross-context clustering

Source peaks are processed by chromosome. A peak can join a cluster only if:

- its context is not already represented in that cluster;
- the complete span of member summits remains at most 150 bp by default; and

- it reciprocally contains a summit with at least one current member: each of the two peak intervals contains the other's summit.

When multiple clusters are eligible, distance to the cluster median summit, then interval-envelope width, gives a deterministic choice. The representative summit is the observed member summit closest to the median; ties prefer a narrower source peak, then stable context/name order.

Adjacent clusters whose representative summits are less than 50 bp apart are merged only when their context sets are disjoint and the combined summit span still satisfies the 150-bp limit. A shared context is evidence that the same biological condition resolves two sites, so those sites remain separate.

Each master interval initially spans the envelopes of its members. If adjacent master intervals overlap, their boundary is clipped at the midpoint between their representative summits, preserving both summits and yielding sorted, non-overlapping, variable-width BED6 intervals.

10.3 Master outputs

The bundle contains the master BED, one-base summit BED, full source-peak membership table, binary context matrix, and JSON metrics. The membership table records input and clipped coordinates, source summit and signal, fallback status, peak method, replicate support, and supporting libraries. A manifest records paths and SHA-256 hashes for the complete bundle.

The summit is a sharper cross-context anchor than a peak edge, while preserving source envelopes avoids pretending that every DHS has the same width. Reciprocal summit containment reduces accidental joins between nearby broad peaks. The one-peak-per-context constraint and shared-context separation rule protect biologically resolved neighboring sites from being collapsed.

11 Strict artifact reuse and validation

Final-BAM and master manifests are immutable contracts between phases. Reuse validates file presence, SHA-256, BAM/BAI consistency, genome, coordinate compatibility, expected filtering metadata, QC status, and master bundle structure before downstream work. Rejected libraries remain visible in the reviewed manifest and exclusion provenance, but they do not enter master or activity calculations.

The result namespace `results/<project>/<run_id>` is protected by a timestamp-independent scientific semantic digest. If an existing namespace contains a different digest, execution stops and requires a new run ID. Stopping stage and report-only inputs are excluded from that digest, so later targets can be added without falsely defining a new scientific analysis.

Cross-run reuse is powerful only when identity is proven. A manifest makes the intended artifact set explicit; hashes prevent path-stable content changes; and the semantic guard prevents two parameterizations from being written into one scientific namespace.

12 Activity quantification

12.1 Assay-specific units

The activity phase requires at least one accepted ATAC and one accepted H3K27ac library for every context, and at least two accepted libraries of each assay across the atlas for TMM estimation.

Paired-end ATAC
: Retain proper pairs with $0 < |\text{TLEN}| < 150$, apply Tn5 offsets, and emit both mates as sorted one-base insertion units.

Paired-end H3K27ac
: Retain one positive-TLEN record per proper pair and emit the inferred genomic fragment once.

Single-end H3K27ac
: Convert filtered alignments to BED and extend in the strand direction using the manually reviewed fragment length, clipping to chromosome bounds.

Each unit BED has a separately checked total count. This same unit definition feeds master-interval counts, 10-kb background counts, H3K27ac summit windows, and normalized browser coverage. The browser representation expands ATAC insertion units to 150-bp intervals, as described below, without changing the one-base units used for counting and normalization.

ATAC accessibility is naturally measured by insertion events, whereas H3K27ac enrichment is naturally measured by immunoprecipitated fragments. Using one internally consistent unit per assay prevents mate double-counting and keeps tables and browser tracks comparable within an assay.

12.2 Exact variable-width master counts

For each library, sorted assay units are intersected with the strict, non-overlapping master registry. Every master DHS receives a raw count and an unnormalized CPM/kb value:

$$\text{CPM/kb}_{i\ell} = \frac{10^9 C_{i\ell}}{N_\ell W_i},$$

where $C_{i\ell}$ is the raw count for element i in library ℓ, N_ℓ is the total number of usable assay units, and W_i is the exact master width in bases. Coordinates, summit, library, assay, context, raw count, total units, and CPM/kb are retained together.

12.3 Assay-specific background TMM

The reference autosomes are tiled into fixed 10-kb bins, with the terminal bin on a chromosome clipped to its length. Raw assay units are counted in every bin. For ATAC and H3K27ac separately, bins with zero total count across that assay are removed and edgeR::calcNormFactors estimates TMM factors using the original total filtered assay-unit counts as library sizes. The implemented parameters are log-ratio trim 0.30, sum trim 0.05, weighting on, and `Acutoff=-1e10`.

For library ℓ, the effective library size is

$$E_\ell = N_\ell F_\ell,$$

where F_ℓ is the assay-specific TMM normalization factor. The normalized element signal is

$$S_{i\ell} = \frac{10^9 C_{i\ell}}{E_\ell W_i}.$$

The factor table records the assay, context, total units, nonzero background feature count, factor, effective size, and method. A software receipt records the R and edgeR versions and exact parameters.

Estimating compositional factors on master DHS counts risks using the biological signal of interest as the normalization reference. Fixed autosomal background bins provide a broader sampling frame. Factors remain assay-specific because insertion counts and ChIP fragments have different generative processes and should never share a scale factor.

12.4 Context aggregation and combined activity

Within each context and assay, normalized biological-library values are combined by an equal-weight arithmetic mean. Replicate standard deviations, raw sums and means, total-unit sums, library IDs, and library counts remain in the output. This gives every accepted biological library equal weight regardless of its number of technical runs.

For context (c) and master DHS (i), the comparative activity score is

$$A_{ic} = \sqrt{\overline{S}_{ic}^{\mathrm{ATAC}}\ \overline{S}_{ic}^{\mathrm{H3K27ac}}}.$$

This geometric mean is zero when either assay has zero signal and balances two positive signals without allowing the numerically larger assay to dominate as strongly as an arithmetic sum.

13 H3K27ac summit-window signal

Master intervals are variable width, but H3K27ac may be displaced to either side of an accessible summit. The catalog therefore defines three non-overlapping nominal 500-bp windows around every representative summit:

$$\begin{aligned} \text{left} &: [s - 750,\ s - 250), \\ \text{center} &: [s - 250,\ s + 250), \\ \text{right} &: [s + 250,\ s + 750). \end{aligned}$$

Windows are clipped at chromosome bounds and their actual widths are retained. H3K27ac fragments are counted in each window, normalized with the same library-specific background-TMM effective sizes, and averaged across biological libraries within the context. The pipeline then selects the maximum of the three context means. It also retains all window-level values and the mean over the complete 1.5-kb neighborhood.

A single window centered on the Tn5 summit can miss flanking nucleosomes, while one wide window can dilute a localized signal or merge asymmetric patterns. The maximum of three fixed local summaries accommodates modest displacement without changing the master coordinate or optimizing a different window for every library.

14 Guarded H3K27ac mixture model

For each context, model fitting uses only master DHSs that are members of that context and have positive max-window H3K27ac signal. Values are transformed to $\log_{10}$ scale. A deterministic expectation–maximization fit compares one Gaussian with a two-Gaussian mixture. The lower-mean and higher-mean components are labeled `low` and `high`.

A two-component interpretation is marked supported only when all of the following guards pass:

1. at least 200 positive member DHSs;
2. $\mathrm{BIC}_1 - \mathrm{BIC}_2 \geq 10$;
3. Ashman's $D \geq 2$;
4. each component weight is at least 0.10;
5. exactly one posterior-probability 0.5 crossing lies between the two component means; and
6. the fitted population has nonzero variance.

If a two-component fit exists, positive member DHSs are assigned using a high component posterior threshold of 0.5 even when one of the support guards fails. Those assignments are retained with `mixture_supported=0`, `mixture_guardrail_warning=1`, and the exact semicolon-separated failure reasons. Nonmembers are `inactive_in_context`; accessible members with zero H3K27ac are `accessible_no_h3k27ac_signal`.

The catalog additionally reports a within-context log-signal z score and descriptive z tiers. These do not replace the mixture assignment.

A two-Gaussian model will return two components even for many unimodal or weakly structured distributions. The guardrails turn model diagnostics into explicit evidence requirements. Retaining unsupported assignments with warnings is preferable to silently discarding data or falsely presenting the fit as validated.

15 Regulatory classes and posterior activity

The representative summit is compared with the nearest reference TSS on the same contig. Ties retain all tied TSS identifiers. Classes are descriptive:

Class	Absolute summit–TSS distance
`promoter_associated`	≤ 250 bp
`proximal_enhancer_like`	251–1,000 bp
`distal_enhancer_like`	$> 1{,}000$ bp
`unclassified_no_tss_on_contig`	no annotated TSS on contig

For each positive-signal context member, the catalog reports `mixture_high_posterior_probability`, the fitted probability of the high H3K27ac component. The `high`/`low` component and the binary `mixture_high_posterior` field are derived at probability 0.5 for summaries; they do not filter the main catalog or the context-element exports. Unsupported fits remain visibly flagged in tables, BED names, plots, summaries, and the final report. A context member with zero H3K27ac has a blank posterior and is labeled `accessible_no_h3k27ac_signal`.

TSS distance provides a reproducible structural annotation, not proof of enhancer function or target-gene identity. Likewise, a high H3K27ac component is an encyclopedia-style activity label. The continuous measurements are the appropriate input for later quantitative scoring and uncertainty-aware work.

16 Catalog tables, tracks, and IGV sessions

16.1 Tabular catalog

The long catalog contains one master-DHS/context row. It includes master coordinates, membership, nearest-TSS distance and identifiers, regulatory class, blacklist overlap flag/base count/fraction, normalized ATAC, all H3K27ac window summaries, mixture diagnostics and posterior, activity state, and the max-window geometric activity. The wide catalog contains one row per master DHS with the TSS and blacklist annotations shared once and the membership, signal, mixture, warning, and activity fields prefixed by context. Each context also receives a compressed table containing every context member, including low-component and zero-signal elements, plus a regulatory-class/mixture summary.

These columns make downstream definitions explicit. For example, an analysis can require `blacklist_overlap=0`, choose a high-component posterior threshold such as 0.9, and apply its own absolute `nearest_tss_distance_bp` boundary. The pipeline does not claim that one such threshold is universally appropriate.

16.2 Portable BED and BigWig tracks

The catalog creates:

- a portable copy of the global master BED;
- a context DHS BED containing master coordinates for context-matrix members, rather than the wider pooled source-peak boundaries;
- a context-element BED9 containing every context member. Its thick one-base interval marks the summit, score is the high-component posterior scaled to 0–1,000 (zero where no posterior

applies), item RGB encodes the regulatory class except that blacklist overlaps are black, and the name records TSS distance, posterior, component, guardrail status, and blacklist status; and

- one mean ATAC and one mean H3K27ac BigWig for every context.

For each accepted library, basewise coverage is scaled by $10^6/E_\ell$, using the same assay-specific background-TMM effective size as the tables. Before ATAC browser coverage is calculated, every shifted one-base insertion is expanded to a centered, chromosome-clipped 150-bp interval. Context tracks are arithmetic means of the scaled library pileups. This makes accessible regions easier to inspect than isolated one-base events, while retaining the insertion-based library definition and the same 150-bp representation used for the MACS3 branch. H3K27ac BigWigs show normalized fragment coverage without additional extension. Per-library bedGraphs exist only as temporary construction files; JSON sidecars record libraries, factors, scale values, representation, inputs, and output checksums.

16.3 IGV sessions

Each context session at `activity/catalog/igv/<context>.xml` loads five relative resources: the global master registry first, followed by the mean 150-bp ATAC pileup, mean H3K27ac, context DHSs, and posterior-scored candidate elements. The genome gene track is enabled, so the master registry is the first custom track immediately below the gene/GTF annotation. The aggregate `activity/catalog/all-contexts.igv.xml` loads the master once and the four context-specific tracks for every context. Relative paths make the bundle portable as long as the XML, `bed/`, and `tracks/` remain together.

The standalone `src/build_igv_session.py` builds a compact cross-run view. The global master DHS registry is the first custom track below the gene/GTF annotation. Each context then contributes, in order, its background-TMM mean 150-bp ATAC Tn5 pileup, pooled ATAC qpois signal, background-TMM mean H3K27ac fragment coverage, context DHSs, and posterior-scored candidate cCREs. The unscaled, depth-dependent pooled MACS3 pileup remains an auditable master-calling result but is intentionally absent from this compact comparative session. Replicate-level H3K27ac signal, H3K27ac peaks, and intermediate ATAC peak evidence are deliberately omitted.

The tabular catalog is the quantitative record; the browser bundle is a diagnostic view of the same accepted libraries, units, factors, and coordinates. Generating both from shared inputs minimizes the risk that a visually attractive track represents a different normalization or sample selection.

17 Context contacts and candidate genes

17.1 Starting from a completed catalog

A completed context-resolved catalog can be used as the input boundary for a new links-only result namespace. The command `src/create\_catalog\_manifest.py` writes a one-row TSV that binds the long catalog, catalog metrics, catalog provenance, and source resolved configuration by SHA-256. The manifest also records the source project, run, method, genome, context order, and semantic digest carried by the source configuration.

At launch, the pipeline verifies all four files and their internal catalog hash, requires the supplied accession table to match the sample-sheet hash recorded by the source catalog, and requires the canonical nine-context dm6 set. The link rules stream the long table and select only rows belonging to the requested context. Quantification, mixture fitting, catalog construction, BigWig generation, and IGV-session generation are not placed in this DAG. This makes the catalog a downstream artifact boundary without assigning the current BAM filtering contract to older alignments that were never processed under it.

Some legacy catalogs predate the `blacklist_overlap` column. The link stage does not rewrite those catalogs; it supplies zero as a link-layer dummy value so that their graph tables retain the current schema. The `blacklist_overlap_annotation` field in each context metrics file labels this case as `legacy_catalog_default_zero`. The dummy means that overlap is unknown, not that the element was checked against the blacklist and found clear.

17.2 Context mapping and source acquisition

The link stage is enabled automatically only when the activity configuration contains the complete canonical nine-context dm6 atlas. Its reviewed source table is `resources/atlas_contact_sources.tsv`, with structure documented by `schemas/contact-sources.schema.yaml`. Restricting the automatic mapping to that exact context set prevents an unrelated catalog from silently inheriting a biologically inappropriate contact experiment.

Context	Assay/strategy	Target resolution	Match represented by the source
`ab`	Micro-C	5 kb	Two adult-CNS biological replicates
`e5`	Micro-C	5 kb	Nuclear-cycle-14 whole embryos, used as the closest stage-5 proxy
`e11`	Micro-C	5 kb	Whole-embryo stages 10–12
`ead`	Micro-C	5 kb	Eye-disc control matrix supplied as a four-sample merge
`lb`	Micro-C	5 kb	Two third-instar larval-CNS biological replicates
`o`	Hi-C	4 kb	Two whole-ovary stage-1–8 maps; assay and stage-range caveats are retained
`wid`	Micro-C	5 kb	Wing-disc WT matrix supplied as a four-sample merge
`e13`	powerlaw	5 kb	No defensible exact whole-embryo map; distance model only
`hid`	powerlaw	5 kb	No defensible exact haltere-disc map; distance model only

The observed matrices are resumably downloaded to `data/raw/contacts/`. The manifest can carry an upstream MD5 or SHA-256 when one is published. Empty checksum cells mean that no source checksum was claimed, not that integrity is ignored: the computed SHA-256 of every source file is frozen in the normalization metrics and final provenance. Source matrices are immutable inputs. GEO supplementary files use one connection; its server rejects the parallel range requests used by the general downloader. The per-file rules declare one `contact_download_slots` unit, allowing the launcher to limit simultaneous source downloads independently of CPU cores.

17.3 Resolution harmonization and balancing

Each multiresolution Cooler is opened at the finest stored resolution that exactly divides the configured target. Single-resolution Coolers and converted HiCExplorer H5 matrices must satisfy the same divisibility rule. Coarsening sums raw pixel counts; approximate coordinate remapping is rejected. Within a context, prepared replicate matrices are then summed by pixel and one ICE normalization is applied to the merged count matrix. The first attempt has a 200-iteration limit. If it does not converge, the calculation restarts from uniform weights with a 2,000-iteration limit. This order gives each read pair one contribution before normalization and avoids averaging independently balanced matrices with incompatible scales or masks.

The retained result is `activity/links/contacts/<context>.balanced.cool`. Its JSON sidecar records source IDs and hashes, original and target resolutions, replicate count, matrix dimensions, the finite-weight fraction, both balancing attempts when a retry was needed, final convergence status, and biological caveats. Conversion and coarsening copies under `work/<project>/<run_id>/activity/contacts/` are reproducible intermediates and are removed after a successful normalized matrix is written; the downloaded source and balanced result are never modified in place.

17.4 Contact-decay model

For every observed context, the workflow extracts finite nonnegative balanced cis contacts from 40 log-spaced diagonal offsets up to 1 Mb and takes the mean of all finite nonnegative pixels, including valid zero pixels, at each usable offset. Masked bins are excluded. A linear fit in log–log space estimates

$$E_c(d) = s_c \max(d, r_c)^{\gamma_c},$$

where r_c is matrix resolution. The floor at one bin avoids an undefined zero-distance expectation. Exponents outside $(-3, -0.05)$ are rejected as implausible instead of being used silently. The atlas model used for `e13` and `hid` takes the arithmetic mean of observed-context exponents and the geometric mean of their scales. Both the individual fits and atlas parameters are retained in `activity/links/contacts/dm6_powerlaw.json`.

This model supplies an explicit distance prior when direct evidence is absent; it does not impute loops or pretend that the two modeled contexts have measured contacts. Every downstream row retains `contact_strategy=powerlaw`, `contact_assay=distance_model`, and the match caveat.

17.5 Promoter nodes and activity

The current dm6 GTF supplies one promoter node for every distinct gene/TSS/strand combination. Alternative transcripts sharing a TSS are collapsed into that node while their transcript IDs are retained. Promoter windows are 500 bp centered on the zero-based TSS and clipped at chromosome boundaries.

For a given context, all context-member master DHSs overlapping a promoter window contribute to its annotation. The node records their IDs, maximum ATAC signal, maximum H3K27ac high-component posterior, and maximum geometric activity $A_j = \sqrt{\mathrm{ATAC}_j \times \mathrm{H3K27ac}_j}$. It is marked active when at least one context-member DHS overlaps and the maximum posterior is at least 0.5. The continuous promoter activity score is

$$P_p = \max_{j \in p} \left[A_j \Pr(\text{high H3K27ac} \mid j) \right].$$

Taking the product within each DHS prevents the maximum activity and maximum posterior from two different overlapping sites from being combined. Using both accessibility and acetylation distinguishes the user's motivating case—a contacted, active promoter from a contacted but inactive promoter— while preserving the underlying values for later thresholds.

17.6 Edges, gene projection, and interpretation

Every context-member master DHS is connected to each annotated promoter on the same chromosome whose TSS lies within 1 Mb of the representative summit. For an observed context, the edge records the ICE-balanced pixel O, the fitted expectation $E_c(d)$, and $O/E_c(d)$. Its contact weight is

$$W_{i,p} = O_{i,p} + 0.01 E_c(d_{i,p}).$$

The small distance-aware pseudocount gives zero pixels a finite, explicitly weak weight. When both anchors occupy one contact bin, the matrix cannot resolve their internal contact; the maximum adjacent-bin value is used and the row is labeled `adjacent_bin_proxy`. For a modeled context, $W_{i,p} = E_{\text{atlas}}(d_{i,p})$, observed contact and enrichment remain blank, and the row is labeled `distance_model`.

The candidate score is

$$S_{i,p} = W_{i,p} P_p.$$

The complete edge table keeps inactive promoters and zero observed pixels; it is not a filtered positive-only network. It stores source/target node IDs and edge evidence without repeating coordinates, transcript IDs, and element annotations on millions of rows; those attributes live in the node table. Alternative promoters are collapsed to one element–gene candidate by the best score. Each gene row includes the best promoter, best active promoter when present, minimum distance, contact and promoter-activity measurements, and four within-element ranks: combined candidate score, active-promoter candidate score, contact-only weight, and nearest-gene distance. These parallel ranks expose why a gene is prioritized.

The focused `<context>.active_contact_enhancer_gene_candidates.tsv.gz` projection is constructed from the promoter edges before gene collapse. It retains only proximal- or distal-enhancer-like DHSs with element H3K27ac high-component posterior at least 0.5 and only edges with observed/expected contact at least 1. Alternative promoters are then collapsed and ranks are recomputed within the retained genes for that enhancer. Promoter activity remains an explicit annotation rather than an additional hidden filter. The output contains only its header for `e13` and `hid`, because a distance prior does not provide an observed/expected measurement. This projection is a separate rule whose only scientific inputs are the completed node and promoter-edge tables. It can therefore be added to a completed links run without downloading or normalizing contacts, refitting decay, or querying the contact matrices again. Its JSON sidecar records both input hashes, the two thresholds, retained element and gene counts, and the output hash.

For the distance-only contexts, a second focused projection is written as <context>.active_distance_enhancer_gene_candidates.tsv.gz. It uses the same enhancer posterior threshold, retains only promoters marked active in that context, collapses alternative promoters to genes, and ranks them by $E_{\text{atlas}}(d)P_p$. The rank-1 row is marked as the primary distance candidate. Each row also reports the closest active promoter TSS and the closest annotated promoter TSS, including ties, which makes the simple nearest-TSS baseline visible beside the activity-weighted distance ranking. Observed contact and observed/expected fields remain blank, and evidence_type=distance_model_active_promoter prevents these rows from being interpreted as measured contacts. An eligible enhancer with no active promoter inside 1 Mb has no candidate row and is counted in the metrics sidecar. Only e13 and hid receive this product. Like the observed-contact projection, it streams completed node and edge tables without repeating contact normalization or graph construction.

The graph products per context are contexts/<context>.nodes.tsv.gz, <context>.element_promoter_edges.tsv.gz, and <context>.element_gene_candidates.tsv.gz, together with the focused active-contact enhancer–gene table and, for distance-only contexts, the active-distance table below activity/links/. They represent candidate relationships. ICE values, the 1calibrated interaction probability, and a high rank is not proof of regulation.

18 Integrated reporting and provenance

The report endpoint emits deterministic HTML, PDF, and JSON. Repeated -report-source-root arguments freeze a bounded set of upstream manifests, resolved configurations, QC metrics, TSS plots, fragment histograms, cross-correlation files, MultiQC reports, and master metrics into the report configuration. These sources are checked again by SHA-256 during report construction.

The integrated report summarizes input/output inventories, BAM and FRiP QC, ATAC TSS and fragment evidence, ChIP cross-correlation, master statistics, TMM factors, regulatory classes, mixture fits, exact guardrail warnings, and the catalog browser assets. When contacts are configured, it also reports the observed-versus-modeled contexts, active promoter counts, and edge/candidate counts. Its JSON sidecar records every declared source and current output hash, including BEDs, mean BigWigs, IGV sessions, and contact-link provenance.

18.1 Reproducibility mechanisms

- Rule software is isolated in small pinned Conda environments.
- Downloads, reused BAMs, master bundles, report sources, and generated exports are checksummed.
- Resolved configurations preserve derived defaults and provenance.
- Temporary outputs are validated and atomically promoted where practical.

- Deterministic tables use stable ordering; generated gzip files set a fixed modification time; unchanged manifests and provenance files are not replaced merely to refresh timestamps.
- Snakemake targets express completeness and profiles retain `rerun-incomplete: true`.
- A changed scientific semantic digest requires a new `run_id`.

The run-specific `work/<project>/<run_id>/` tree contains reproducible intermediates rather than archive products. Large intermediates—trimmed FASTQs after alignment, lane and marked BAMs, shifted or short-fragment BAMs, peak-calling insertion BEDs and background-count tables—are declared temporary and Snakemake removes them after their last consumer. Quantification assay-unit BEDs are retained below `activity/quantification/units/`; they are the strict restart contract for catalog counting and mean-track construction. The work directory must remain available while downstream products are being constructed. After successful completion it may be removed to clear small files or debris left by interrupted older runs. Per-library ChIP CPM BigWigs are not computed because no downstream result consumes them; pooled ATAC pileup/qpois and context-mean catalog BigWigs are distinct retained products. Final replicate-level ATAC peak evidence is archived below `results/<project>/<run_id>/atac/replicates/` rather than retained under `work/`.

19 Principal output inventory

Group	Principal paths below `results/<project>/<run_id>/`
Alignment	`bam/<library>.final.bam[.bai]`; `qc/alignment/`; `provenance/manifests/final-bams.tsv`
QC and review	`qc/metrics.tsv`, `qc/metrics.json`, `qc/library-review.tsv`, `qc/multiqc/multiqc_report.html`, assay-specific files below `qc/tss/`, `qc/fragments/`, and `qc/chip/`
Context ATAC	`atac/conditions/<context>/peaks/` and `atac/conditions/<context>/tracks/`
Master bundle	`atac/master/master_dhs.bed`, `master_dhs_summits.bed`, `master_dhs_membership.tsv`, `master_dhs_context_matrix.tsv`, and `master_dhs.json`
Quantification	`activity/quantification/units/*.bed.gz`, `activity/quantification/libraries/*.signal.tsv.gz`, `normalization_factors.tsv`, `context_signal.tsv.gz`, `master_dhs_activity.tsv.gz`, metrics and provenance JSON
Catalog	`activity/catalog/master_elements_long.tsv.gz`, `master_elements_wide.tsv.gz`, `elements/`, `mixture_models.tsv`, `regulatory_element_summary.tsv`, metrics, provenance, and mixture plots
Browser bundle	`activity/catalog/bed/`, `activity/catalog/tracks/`, `activity/catalog/igv/<context>.xml`, and `activity/catalog/all-contexts.igv.xml`

Group	Principal paths below `results/<project>/<run_id>/`
Contact links	`activity/links/contacts/*.balanced.cool`, `contacts/dm6_powerlaw.json`, `promoters.tsv.gz`, `contexts/*.nodes.tsv.gz`, `*.element_promoter_edges.tsv.gz`, `*.element_gene_candidates.tsv.gz`, `*.active_contact_enhancer_gene_candidates.tsv.gz`, and the distance-context `*.active_distance_enhancer_gene_candidates.tsv.gz` with metrics sidecars, aggregate metrics, and provenance JSON
Integrated report	`activity/report/integrated_qc_report.html`, `integrated_qc_report.pdf`, and `integrated_qc_report.json`
Restart contracts	`provenance/manifests/<stage>.checkpoint.json` at every logical endpoint

20 Representative commands

20.1 Accessions through QC

```
python src/run_pipeline.py resources/atlas_samples_ip_only.tsv \
  --project drosophila-atlas --run-id qc-v1 --genome dm6 \
  --until-stage qc --cores 16
```

Copy `qc/library-review.tsv` outside the result namespace, fill the final review columns, and apply the decisions:

```
python src/review_final_bam_manifest.py \
  results/drosophila-atlas/qc-v1/provenance/manifests/final-bams.tsv \
  --review-table data/reviewed/atlas.library-review.tsv \
  --output data/reviewed/atlas.reviewed.tsv
```

20.2 Reviewed ATAC BAMs through master

```
python src/run_pipeline.py resources/atlas_samples_ip_only.tsv \
  --from-stage qc \
  --checkpoint-manifest \
    results/drosophila-atlas/qc-v1/provenance/manifests/qc.checkpoint.json \
  --final-bam-manifest data/reviewed/atlas-atac.reviewed.tsv \
  --project drosophila-atlas --run-id master-v1 --genome dm6 \
  --until-stage master --cores 16
```

20.3 Accepted BAMs and master through report

```
python src/run_pipeline.py resources/atlas_samples_ip_only.tsv \
  --from-stage master \
  --master-manifest \
    results/drosophila-atlas/master-v1/provenance/manifests/master-dhs.tsv \
```

```
  --activity-bam-manifest data/reviewed/atac.accepted.tsv \
  --activity-bam-manifest data/reviewed/h3k27ac.accepted.tsv \
  --report-source-root results/drosophila-atlas/master-v1 \
  --report-source-root results/drosophila-h3k27ac/qc-v1 \
  --project drosophila-atlas --run-id catalog-v1 --genome dm6 \
  --until-stage report --cores 16
```

20.4 Completed catalog through contact links

```
python src/create_catalog_manifest.py \
  results/drosophila-atlas/catalog-from-reviewed-v1 \
  --output data/reviewed/atlas-catalog.reviewed.tsv

python src/run_pipeline.py resources/atlas_samples_ip_only.tsv \
  --from-stage catalog --until-stage links \
  --catalog-manifest data/reviewed/atlas-catalog.reviewed.tsv \
  --project drosophila-atlas \
  --run-id contact-links-from-catalog-v1 --genome dm6 \
  --cores 8 \
  --snakemake-arg=--resources \
  --snakemake-arg=mem_mb=16000 \
  --snakemake-arg=contact_download_slots=2
```

This command creates only `activity/links/` and links-stage provenance in the new run. The original catalog remains immutable and is referenced by its manifest and hashes.

Re-running the same command and run ID resumes valid work. Use `quantification`, `catalog`, or `links` as an earlier endpoint. For the canonical nine-context dm6 atlas, choosing `links` or `report` downloads and normalizes the versioned contact sources. To advance later, pass an exported checkpoint as the completed start boundary:

```
python src/run_pipeline.py resources/atlas_samples_ip_only.tsv \
  --from-stage quantification \
  --checkpoint-manifest \
    results/drosophila-atlas/catalog-v1/provenance/manifests/quantification.checkpoint.
    json \
  --until-stage report --cores 16
```

21 Default parameters that affect interpretation

Parameter	Default	Interpretation
Trim quality / minimum length	20 / 20 bp	Read-level cleanup before alignment
Bowtie2 preset / paired maximum	very-sensitive / 2,000 bp	Alignment sensitivity and allowed concordant span
MAPQ / duplicate / mitochondrial filtering	30 / remove / remove	Final biological-library BAM contract; blacklist alignments are retained

Parameter	Default	Interpretation
Blacklist handling	annotate only	Master-DHS overlap flag, base count, and interval fraction
ATAC short-fragment maximum	150 bp, strict	Requires $0 < \lvert\mathrm{TLEN}\rvert < 150$
ATAC browser extension	150 bp	Centered Tn5 insertion pileup recorded in the resolved semantic configuration
ATAC MACS3 q-value	0.10	Lenient candidate stage before geometric refinement
qpois exponent range	2–325	Progressive signal thresholds
qpois retained width	50–400 bp	Final refined component geometry
Context replicate support	2 libraries, 0.5 coverage	Minimum reproducibility of pooled peaks
Master summit span	150 bp	Complete-linkage limit across context summits
Minimum master summit separation	50 bp	Close disjoint-context clusters may merge below this distance
TMM background bins	10 kb autosomal	Composition-factor feature space
H3K27ac windows	three 500-bp windows	Left, center, and right of the master summit
Mixture minimum population	200 positive members	Small contexts cannot support the bimodality claim
Mixture evidence	$\Delta\mathrm{BIC} \geq 10$, $D \geq 2$, weights ≥ 0.10	Required together with unique crossing and nonzero variance
Promoter/proximal boundaries	250 / 1,000 bp	Absolute summit-to-nearest-TSS distance
Contact promoter window	500 bp	Centered on each distinct dm6 GTF gene/TSS node
Maximum link distance	1 Mb	Same-chromosome element summit to promoter TSS
Observed contact pseudocount	1% of fitted expectation	Finite weak weight for a zero matrix pixel
Promoter active posterior	0.5	Requires an overlapping context-member DHS and H3K27ac high-component posterior
Focused enhancer posterior	0.5	Element-level threshold for both focused gene tables
Focused observed/expected contact	1.0	Applied to observed promoter edges before gene collapse
Focused distance promoter activity	Active	Required for distance-only gene candidates

22 Pinned principal software

The rule environments pin the principal tools shown below. The resolved Conda environment and report receipts remain the authoritative record for an individual run.

Tool	Version	Tool	Version
Snakemake	9.23.1	Bowtie2	2.5.5
samtools	1.23.1	bedtools	2.31.1
FastQC	0.12.1	Cutadapt	5.2
MACS3	3.0.4	deepTools	3.5.6
MultiQC	1.35	phantompeakqualtools	1.2.2
R	4.5.3	edgeR	4.8.2
pyBigWig	0.3.25	WeasyPrint	66.0
pandas	2.2	Cooler	0.10.4
HiCExplorer	3.7.6	HiCMatrix	17.2

23 Interpretive cautions

- A refined qpois component inherits its candidate call; refinement is not a new independent false-discovery-rate procedure.
- Replicate support is interval coverage of a pooled peak, not a test of identical peak boundaries or identical signal magnitude.
- A master DHS represents compatible accessibility evidence across contexts. It is not a fixed-width enhancer, a gene assignment, or a claim that the element is active in every context.
- Background TMM assumes that a useful reference composition can be estimated from autosomal bins. Factor ranges, library QC, and signal distributions should be inspected for strong global or technical shifts.
- A high mixture assignment from an unsupported fit is retained for transparency but must be interpreted with its warning. The report exposes the exact failed guards.
- A blacklist overlap is a technical warning, not proof that an element is artifactual. Conversely, retaining the reads can allow blacklist-driven signal to affect peak calling, so sensitivity analyses should state the overlap rule they apply.
- TSS-proximity classes are annotation categories. Distal or proximal labels do not establish enhancer function, and promoter proximity does not prove regulation of the nearest gene.
- Browser tracks are normalized context means intended for inspection; replicate-level tables and standard deviations remain necessary for uncertainty assessment.
- Contact match quality differs by context. The ovary data are Hi-C, several Micro-C sources are close proxies or pre-merged matrices, and `e13/hid` have distance-only evidence. These labels must be retained in comparative analyses.

- ICE-balanced pixels and the fitted distance prior are relative contact evidence. Candidate scores are useful within-element rankings but are not calibrated probabilities, causal assignments, or directly comparable expression effects across contexts.
- ABC scoring requires a downstream workflow that consumes the continuous signals, links, and frozen master coordinates.

24 Verification of this description

This document was written against the current sample-sheet schema, launcher, configuration validators, `workflow/Snakefile`, all included rule files, the master/consensus/activity/catalog/contact-link implementations, the README, and the workflow environment files. To rebuild the PDF from the repository root:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  -output-directory=docs docs/pipeline-description.tex
```

For a scientific run, documentation review is not sufficient verification. Validate the sample sheet, inspect the resolved configuration and semantic digest, run an appropriate Snakemake dry-run, review library QC decisions, and retain the generated manifests and report JSON with the results.